Stormwater Regulation Summary: Nashville, TN

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Flood control and water quality — Nashville experienced catastrophic flooding in 2010 ($2B+ damage).
- Metro Nashville Stormwater Management Manual (2021) administered by Metro Water Services.
- Applies to projects with ≥10,000 SF cumulative impervious (stricter in special flood hazard and buffer areas).
- Dual requirements: Water Quality (WQ) volume capture + peak flow rate control for multiple storm events.
- NPDES MS4 permit requires post-construction stormwater management for all qualifying developments.

2. Typical Soil Conditions

- Nashville Basin is predominantly limestone karst terrain underlain by Ordovician-age limestone.
- Typical soils: Maury silt loam (well-drained, B), Mimosa-Braxton complex (moderately well-drained, C).
- Hydrologic Soil Group: Mixed B and C — better than many major cities but karst creates unique challenges.
- Karst features (sinkholes, solution channels) are common — infiltration BMPs must avoid directing water into karst voids.
- Implication: While soils can support some infiltration, karst risk pushes many sites toward retention/detention rather than direct infiltration.
- *Karst assessment required for sites in the Inner Basin — consult Metro Nashville sinkhole database.*

3. Green Roof Requirements

- Green roofs are not mandatory — Nashville does not have a green roof ordinance or building requirement.
- Metro Nashville accepts green roofs as an approved BMP for WQ volume management.
- Stormwater fee credit is available for properties that manage runoff on-site, including via green roof.
- TVA Green Invest program supports renewable energy/green infrastructure but does not specifically target green roofs.
- Market is less mature than coastal cities — opportunity to educate engineers and developers on rooftop stormwater management.

4. TSS Requirements

- Nashville requires 80% TSS removal as part of the Water Quality volume management standard (1.0" WQ event).
- Green roofs can contribute to TSS compliance — capturing the WQ event (1.1") addresses the first-flush pollutant load.
- Metro Nashville Stormwater Manual lists green roofs among approved BMPs with pollutant removal credits.
- Additional TSS treatment may be needed for ground-level impervious areas not managed by rooftop BMPs.
- *Specific TSS removal credit percentages for green roofs should be confirmed in current Metro Nashville BMP manual.*

5. Nature-Based Retention Requirements

- Capture and treat the Water Quality Volume (WQv): 1.0" event (preceded by 72-hr dry period) from all impervious surfaces.
- WQv can be managed through retention (infiltration, evapotranspiration) or treatment (filtration, settling).
- CN (Curve Number) method per NRCS TR-55 is the standard for peak flow calculations.
- Typical Nashville urban CN: 80-90 (developed); target with BMPs: 65-75.
- Metro Nashville uses the Critical Storm Method for some detention sizing — verify which method applies to the specific project.

6. Allowable Outflow Rates

- Peak flow control required for 2-yr, 10-yr, 25-yr, and 100-yr storm events.
- Post-development peak flow must not exceed pre-development peak flow for each design storm.
- Nashville does not use a flat l/s/ha standard — control is relative to pre-development conditions.
- For reference, typical undeveloped Nashville site (CN 65-70): approximately 15-30 l/s/ha for the 10-year storm.
- *Site-specific hydrology required; Metro Water Services reviews each project's drainage study.*
- *Special Flood Hazard Areas may have additional flow restrictions beyond standard requirements.*

Items in *italic* should be verified against current municipal codes. | Generated by Sempergreen USA Stormwater BMP Comparison Tool